$$\frac{140+142+150+150+140+148+140+147+145+140+147+145}{12}=\frac{1734}{12}=144.5$$

(ii) You can use Assumed mean method. Assume, 141 as the assumed mean.

Then the mean will be given by

$$=141+\frac{(-1)+(1)+(9)+(9)+(-1)+(7)+(-1)+(6)+(4)+(-1)+(6)+(4)}{12}$$

$$=141+\frac{-4+46}{12}=141+\frac{42}{12}=141+3.5=144.5$$

(iii) A third method is to deal with an ungrouped frequency distribution. You find that 140 has occurred 4 times, (implying 4 is the frequency of 140), 142 has occurred only once (indicating that 1 is the frequency of 142) and so on. This enables us to get the following frequency distribution.

Height(cm)	140	142	150	148	145	147
No. of students	4	1	2	1	2	2

You find that there are four 140s; their total will be $140 \times 4 = 560$

There is only one 142; so the total in this case is $142 \times 1 = 142$

There are two 150s; their total will be $150 \times 2 = 300$ etc.

These details can be neatly tabulated as follows:

Height (x)	Frequency (f)	fx
140	4	560
142	1	142
150	2	300
148	1	148
145	2	290
147	2	294
	12	1734

$$\text{Mean} = \frac{\text{Sum of all } fx}{\text{No.of items}}$$

$$=\frac{1734}{12}=144.5 \text{ cm}$$

Note

Study each step and understand the meaning of each symbol.

Looking at the procedure in general terms, you can obtain a formula for ready use. If $x_1, x_2, x_3, \ldots x_n$ are n observations whose corresponding frequencies are $f_1, f_2, f_3, \ldots f_n$ then the mean is given by

$$\bar{X} = \frac{f_1x_1+f_2x_2+\ldots+f_nx_n}{f_1+f_2+\ldots f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{\sum_{i=1}^{n} f_i} = \frac{\sum fx}{\sum f}$$

Can you adopt the above method combining with the assumed mean method? Here is an attempt in that direction:

8-Stats.indd 271 02/12/2022 21:51:13